Lecture 21, Part II - Lambda switch: Genetic switches

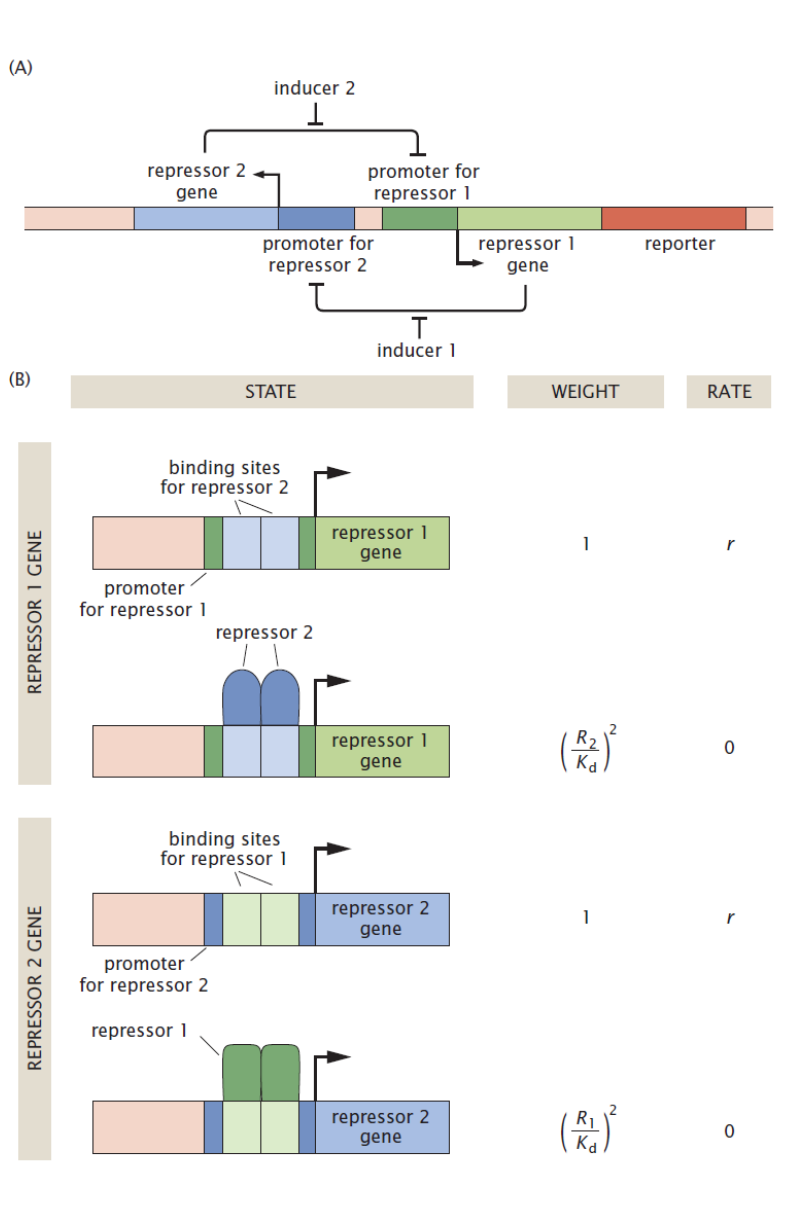
1/20/17 6:20 AM

Outline:

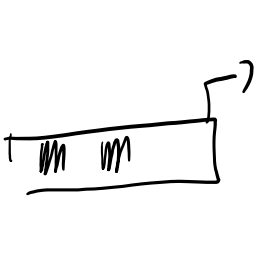
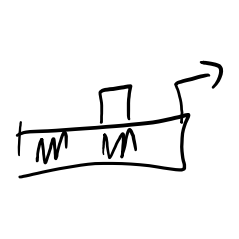
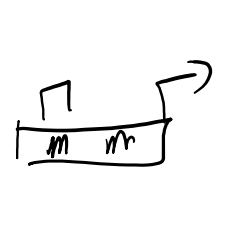
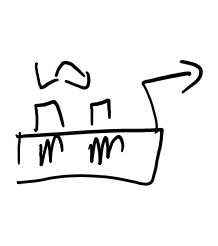
- Switches everywhere
  - ↳ HSC
  - ↳ Fly development
  - ↳ Look @ other reviews
- Two experiments
  - ↳ Harinder
  - ↳ Bacterial switch
- HW: Autoactivation

From binding energies to Kds:

- What is the NS reservoir is the solution
- Calculate pBound in terms of Kd
- Show equivalence of statistica weights



states and weights of the mutual repressor switch w/thermodynamics:

STATE	WEIGHT	RATE
	1	r
	R_1/K_d	0
	R_1/K_d	0
	$\left(\frac{R_1}{K_d}\right)^2 \omega$	0

$$\frac{dR_2}{dt} = \frac{1}{1+2\frac{R_1}{K_d}+\left(\frac{R_1}{K_d}\right)^2 \omega} r - \gamma R_2$$

$$\frac{dR_1}{dt} = \frac{1}{1+2\frac{R_2}{K_d}+\left(\frac{R_2}{K_d}\right)^2 \omega} r - \gamma R_1$$

The nullclines:

$$R_2 = \frac{r}{\gamma} \frac{1}{1+2\frac{R_1}{K_d}+\left(\frac{R_1}{K_d}\right)^2 \omega}$$

$$R_1 = \frac{r}{\gamma} \frac{1}{1+2\frac{R_2}{K_d}+\left(\frac{R_1}{K_d}\right)^2 \omega}$$

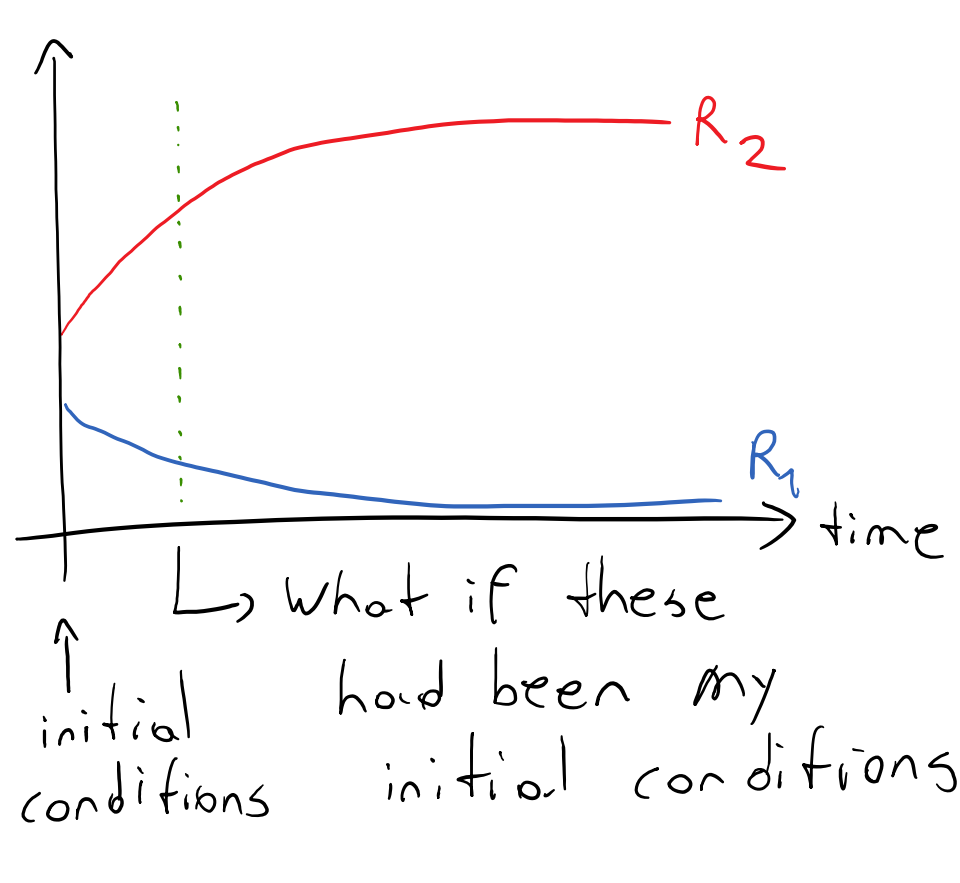
```
%Model parameters:
r=10;      %1/min
gamma=1;   %1/min
Kd=5;      %nM
%Note, it's very interesting to see how the bifurcation emerges as you
%increase w from 1.
w=5;

%Simulation parameters
R1=linspace(0,20);
R2=linspace(0,20);



%Calculate the vector field.
%For-loops that move through each possible value of R1 and R2
for i=1:length(R1)
   for j=1:length(R2)
      VectorR1(j,i)=1/(1+2*R2(j)/Kd+(R2(j)/Kd)^2*w)*r-gamma*R1(i);
      VectorR2(j,i)=1/(1+2*R1(i)/Kd+(R1(i)/Kd)^2*w)*r-gamma*R2(j);
   end
end


figure(2)
plot(R1,r/gamma*1./(1+2*R1/Kd+(R1/Kd).^2*w),'-k')
hold on
plot(r/gamma*1./(1+2*R2/Kd+(R2/Kd).^2*w),R2,'-r')
hold off
xlim([0,10])
ylim([0,10])
legend('dR2/dt=0','dR1/dt=0')
hold on
quiver(R1(1:2:end),R2(1:2:end),VectorR1(1:2:end,1:2:end),...
   VectorR2(1:2:end,1:2:end),'LineWidth',1,'MaxHeadSize',3,...
   'AutoScaleFactor',1.5)
hold off
```

Revisiting the concept of the phase diagram:

$$\frac{dR_2}{dt} = \frac{1}{1+2\frac{R_1}{K_d}+\left(\frac{R_1}{K_d}\right)^2 \omega} r - \gamma R_2$$

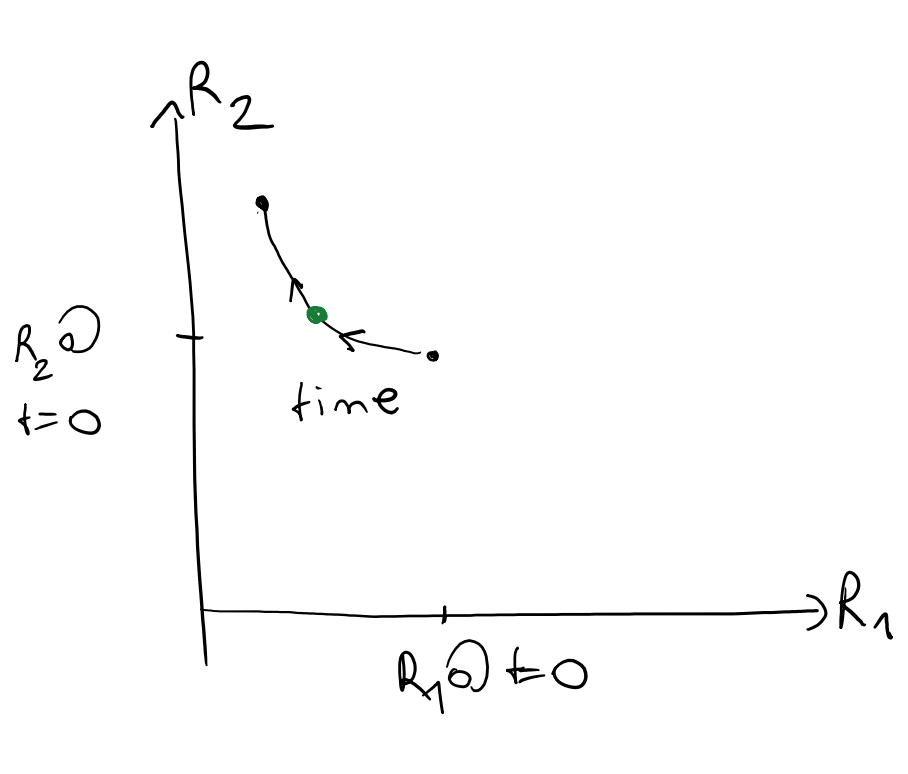
$$\frac{dR_1}{dt} = \frac{1}{1+2\frac{R_2}{K_d}+\left(\frac{R_2}{K_d}\right)^2 \omega} r - \gamma R_1$$

Given initial conditions, I can solve the system dynamics



↳ What if these had been my initial conditions?

↑ initial conditions

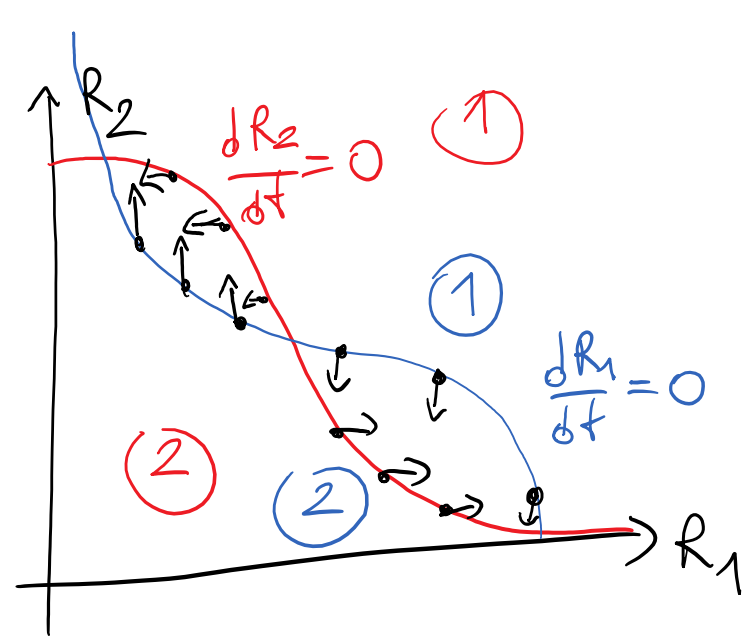


∴ How the system evolves depends on where I start. If I choose a point in the phase diagram as my initial condition, I can calculate $\frac{dR_1}{dt}$ and $\frac{dR_2}{dt}$ and figure out where it will be at the next time point

Nullclines in a phase diagram:

$$R_2 = \frac{r}{\gamma} \frac{1}{1+2\frac{R_1}{K_d}+\left(\frac{R_1}{K_d}\right)^2 \omega}$$

$$R_1 = \frac{r}{\gamma} \frac{1}{1+2\frac{R_2}{K_d}+\left(\frac{R_1}{K_d}\right)^2 \omega}$$



① $R_2 \gamma > r \frac{1}{1+2\frac{R_1}{K_d}+\left(\frac{R_1}{K_d}\right)^2 \omega}$

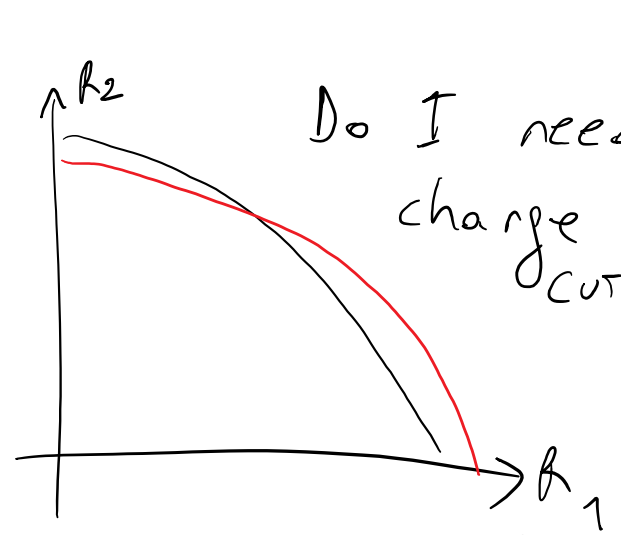
$\Rightarrow \frac{dR_2}{dt} < 0$

② $\frac{dR_2}{dt} > 0$

① $\frac{dR_1}{dt} < 0$

② $\frac{dR_1}{dt} > 0$

Can I figure out when I'll get a bifurcation?



Do I need a change of curvature?

$$\frac{dR_2}{dR_1} = \frac{r}{\gamma} \frac{-\frac{2}{K_d} - 2\frac{R_1}{K_d}\cdot\frac{1}{K_d}\omega}{\left[1+2\frac{R}{K_d}+\left(\frac{R_1}{K_d}\right)^2\omega\right]^2}$$

$$= -\frac{r}{\gamma}\cdot 2\frac{1}{K_d} \frac{1+\frac{R_1}{K_d}\omega}{\left[1+2\frac{R}{K_d}+\left(\frac{R_1}{K_d}\right)^2\omega\right]^2}$$

$$\frac{d^2R_2}{dR_1} = -\frac{r}{\gamma} 2\frac{1}{K_d} \cdot \frac{\frac{\omega}{K_d}\left[1+2\frac{R}{K_d}+\left(\frac{R_1}{K_d}\right)^2\omega\right]^2 - \left(1+\frac{R_1}{K_d}\omega\right)2\left[1+2\frac{R_1}{K_d}+\left(\frac{R_1}{K_d}\right)^2\omega\right]\cdot\left(2\frac{1}{K_d}+\omega 2\frac{R_1}{K_d^2}\right)}{\left[1+2\frac{R}{K_d}+\left(\frac{R_1}{K_d}\right)^2\omega\right]^4}$$

Is there a change in curvature?

$$0 = \frac{\omega}{K_d}\left[1+2\frac{R}{K_d}+\left(\frac{R_1}{K_d}\right)^2\omega\right]^{\cancel{2}} - \left(1+\frac{R_1}{K_d}\omega\right)2\cancel{\left[1+2\frac{R_1}{K_d}+\left(\frac{R_1}{K_d}\right)^2\omega\right]}\cdot\left(2\frac{1}{K_d}+2\omega\frac{R_1}{K_d^2}\right)$$

$$0 = \frac{\omega}{K_d}\left[1+2\frac{R_1}{K_d}+\left(\frac{R_1}{K_d}\right)^2\omega\right] - 4\left(\frac{1}{K_d}+2\omega\frac{R_1}{K_d^2}+\cancel{\frac{R_1}{K_d^2}}\omega+\omega^2\frac{R_1^2}{K_d^3}\right)$$

$$0 = R_1^2\left(\frac{\omega^2}{K_d^3}-\frac{\omega^2}{K_d^3}4\right)+R_1\left(2\frac{\omega}{K_d^2}-8\frac{\omega}{K_d^2}\right)+\left(\frac{\omega}{K_d}-4\frac{1}{K_d}\right)$$

$$0 = R_1^2\left(-3\frac{\omega^2}{K_d^3}\right)+R_1\left(-6\frac{\omega}{K_d^2}\right)+\frac{1}{K_d}(\omega-4)$$

$$R_1 = \frac{6\frac{\omega}{K_d^2} \pm \left[36\frac{\omega^2}{K_d^4}+12\frac{\omega^2}{K_d^3}(\omega-4)\right]^{1/2}}{-6\frac{\omega^2}{K_d^3}}$$

$\overset{\vee}{0}$?

$$36\frac{\cancel{\omega^2}}{K_d^{\cancel{4}}}+12\frac{\cancel{\omega^2}}{\cancel{K_d^3}}(\omega-4) > 0$$

$$36+12\cdot K_d(\omega-4) > 0$$

$$3+K_d(\omega-4) > 0$$

$$K_d(\omega-4) > -3$$

$$\omega-4 > -\frac{3}{K_d}$$

Put this in Mathematica:

$$R_1 = \frac{-3\omega \pm \sqrt{3}\sqrt{(\omega-1)\omega^2}}{3\omega^2}$$

$$= \frac{-3\cancel{\omega} \pm \cancel{\omega}\sqrt{3(\omega-1)}}{3\omega^{\cancel{2}}}$$

$$= \frac{-3 \pm \sqrt{3(\omega-1)}}{3\omega}$$

Do I ever have a positive solution?

$$-3+\sqrt{3(\omega-1)} > 0$$

$$\sqrt{3(\omega-1)} > 3$$

$$3(\omega-1) > 9$$

$$\omega-1 > 3$$

$$\omega > 2$$

It doesn't make sense that things would not depend on $\alpha = \frac{r}{\gamma}\frac{1}{K_d}$

Try again finding the fixed points. In Mathematica, the first two roots correspond to fixed points (if they exist). The third one is $R_1 = R_2$. The other ones are complex numbers.
Let's look at the first root:

> 0?

$$4-4\omega-4\alpha\omega+\alpha^2\omega^2 = 0$$

$$\omega^2\alpha^2+\omega 4(-1-\alpha)+4 = 0$$

$$\omega = \frac{4(1+\alpha) \pm \sqrt{16(1+\alpha)^2-4\alpha^2 4}}{2\alpha^2}$$

$$\omega = \frac{4(\alpha+1) \pm \sqrt{16+32\alpha+\cancel{16\alpha^2}-\cancel{16\alpha^2}}}{2\alpha^2}$$

$$= \frac{4(\alpha+1) \pm 4\sqrt{1+2\alpha}}{2\alpha^2}$$

Adapting this for the transcription book

$$\frac{dR_1}{dt} = \frac{1}{1+2\frac{R_2}{K_d}+\left(\frac{R_2}{K_d}\right)^2 \omega} r - \gamma R_1$$

Let's make it non-dimensional

$$\frac{d R_1/K_d}{dt/\gamma} = \frac{1}{1+2\frac{R_2}{K_d}+\left(\frac{R_2}{K_d}\right)^2 \omega} \frac{r}{K_d \gamma} - \frac{R_1}{K_d}$$

$$\frac{dr_1}{d\tau} = \frac{1}{1+2r_2+r_2^2\omega} r' - r_1$$

$$\frac{dr_2}{d\tau} = \frac{1}{1+2r_1+r_1^2\omega} r' - r_2$$

In S.S.

$$r_2 = \frac{r'}{1+2r_1+r_1^2\omega}$$

$$\Rightarrow r_1(1+2r_2+r_2^2\omega) = r'$$

$$r_1\left[1+2\frac{r'}{1+2r_1+r_1^2\omega}+\left(\frac{r'}{1+2r_1+r_1^2\omega}\right)^2\omega\right] = r'$$

$$r_1\left[\frac{(1+2r_1+r_1^2\omega)^2}{(1+2r_1+r_1^2\omega)^2}+\frac{2r'(1+2r_1+r_1^2\omega)}{(1+2r_1+r_1^2\omega)^2}+\frac{r'^2\omega}{(1+2r_1+r_1^2\omega)^2}\right] = r'$$

$$r'\cdot(1+2r_1+r_1^2\omega)^2 =$$